An Exploratory Analysis of Striking Within the UFC

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This project explores the striking data of all fighters who have competed in the UFC through two connected analyses: plotting volume against accuracy and damage given vs. damage taken.

Main Figures:

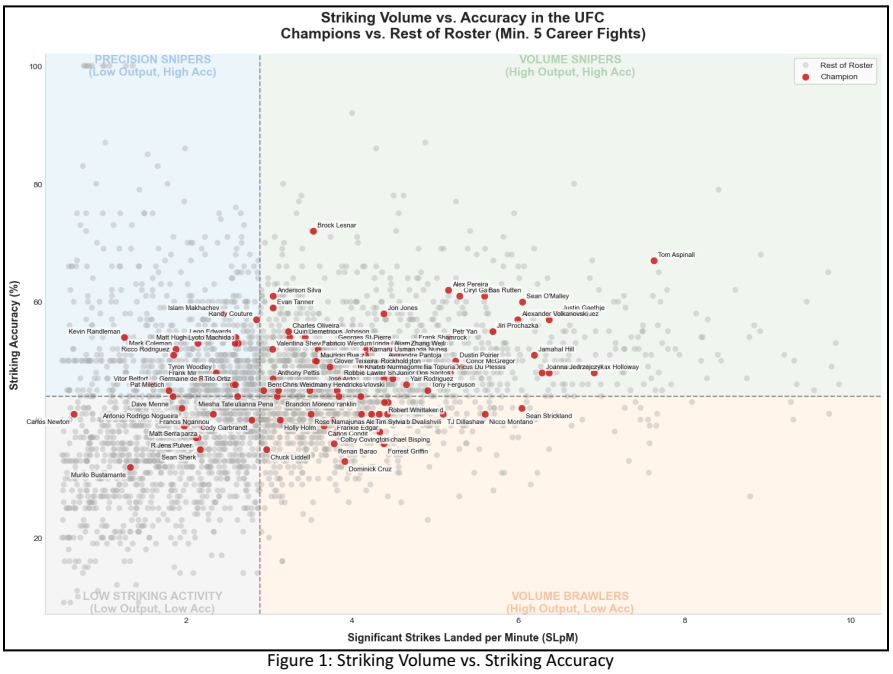

Figure 1: Striking Volume vs. Striking Accuracy

Legend (Figure 1):

Axes: Significant Strikes Landed (SlpM) vs. Accuracy (%)

Grey Dots: Non Champion UFC roster members. **Red Dots & Labels:** UFC Champions.

Dotted Center Lines: Exact median output and median accuracy of the UFC roster.

Green Quadrant: "Volume Snipers" (High Output, High accuracy).

Blue Quadrant: "Precision Snipers" (Low Output, High accuracy).

Orange Quadrant: "Volume Brawlers" (High Output, Low accuracy).

Grey Quadrant: "Low Striking Activity" (Low Output, Low Accuracy).

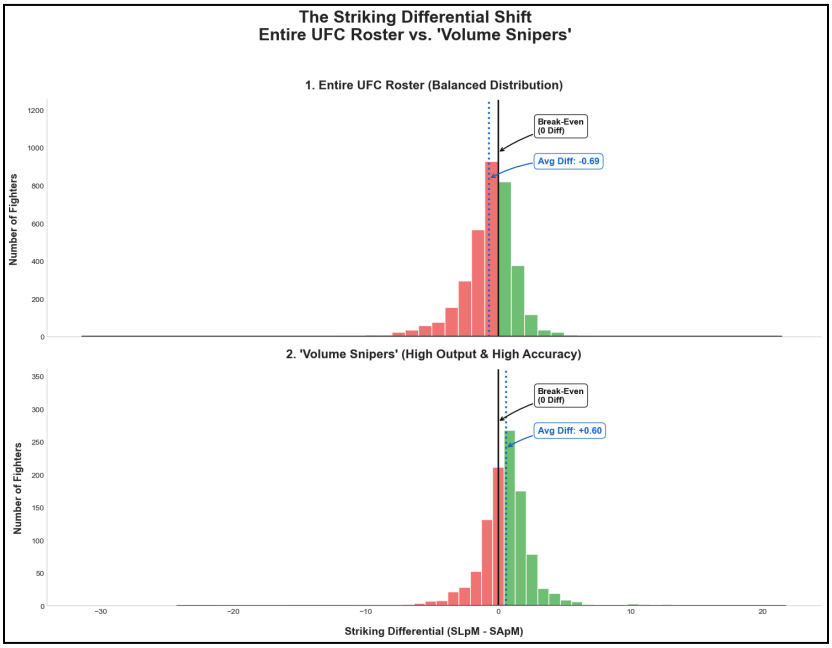

Figure 2: Striking Differential Comparison of Entire UFC Roster and Volume Snipers

Legend (Figure 2):

Axes: Striking Differential (Strikes Landed per Min - Strikes Absorbed per Min) vs. Number of Fighters
Green Bins: Positive Striking Differential (Landing more strikes than absorbing).
Red Bins: Negative Striking Differential (Absorbing more strikes than landing).
Solid Black Line: Break-Even point (0 Differential - one strike landed and one strike absorbed).
Dotted Blue Line: The average striking differential for the respective population.
Dark Grey Curve: Kernel Density Estimate (KDE) showing overall distribution shape.

Findings:

- **"Volume Sniper" Success:** A majority of UFC champions fall into the top-right quadrant of Figure 1 - Volume Snipers (High Output, High Accuracy) which reveals a correlation between success in the UFC and high striking output AND high striking accuracy.
    - As demonstrated in Figure 2, isolating these "Volume Snipers" causes the entire population distribution to shift to the right, raising the average striking differential from -0.69 to +0.60 proving high output paired with high accuracy generally guarantees a net-positive striking exchange.
- **Negative Striking Differential for the Whole Roster:** Seen in the top of Figure 2, the entire UFC roster average striking differential is negative. This statistically confirms that the average fighter takes more damage than they inflict.
    - The average striking differential is -0.69 meaning that, on average, the fighter absorbs 0.69 more significant strikes per minute than they land.
- **Low Striking Activity Champions:** There are champions that are in or near the bottom-left quadrant of Figure 1 -Low Striking Activity (Low Output, Low Accuracy).
    - With too many data points to consider these champions outliers it's important to consider that the sport of MMA is not just striking but grappling as well.
    - Further research into the lowest striking activity champion (Murilo Bustamente), reveals that he is black belt in Brazilian Jiu Jitsu which is a grappling-heavy martial art.

Data & Methods:

Data Source: The dataset "ufc_fighters_final.csv" (source:https://www.kaggle.com/datasets/jossilva3110/ufc-dataset-1994-2026?select=ufc_fighters_final.csv) comprises all-time performance metrics for every UFC fighter from 1994 to 2026. To ensure statistical reliability and exclude irregular fighters, the dataset was filtered to include only those with a minimum of five total career fights.

Variables: The core metrics analyzed are Significant Strikes Landed per Minute (SLpM) and Striking Accuracy (%). The calculated metric, "Striking Differential," was derived by subtracting Strikes Absorbed

per Minute (SApM) from SLpM. A positive differential indicates a fighter deals more damage than they absorb. A negative differential indicates a fighter absorbs more damage than they give out.

Visualizations and Analysis: Python was utilized within a Jupyter Notebook for all data manipulation and visualization. The pandas library was used to compute roster medians, filter the "Volume Sniper" cohort, and matched all UFC Champions. Visualizations were constructed using matplotlib and seaborn. A custom scatter plot was built to map fighting archetypes across four defined quadrants (Figure 1). Stacked histograms overlaid with Kernel Density Estimate (KDE) curves were utilized to demonstrate the continuous probability density of striking differentials across “Volume Snipers” and the entire UFC roster (Figure 2).

Significance Statement:

Historically, success in MMA has been difficult to concretely pinpoint due to the constant evolution of the sport. However, as the sport matures there are greater opportunities when it comes to finding trends. Data visualization reveals the true metrics of championship success as it pertains to striking. By defining distinct fighting archetypes and mapping over 70 UFC champions, this project uncovers vital, objective insights for athletes to optimize their training or even coaches formulating data-backed game plans for fights.

Further, by combining archetype mapping with striking differential analysis, this report visually identifies how elite “Volume Snipers”, UFC Champions, and the average UFC fighter navigate the risk-reward tradeoff of striking. Again, this provides essential insight for coaches designing risk-avoidant fight strategies, sports bettors predicting bout outcomes, and sports health scientists evaluating the longevity of a fighter's career.

Additional Resources Used:

https://medium.com/data-science/how-to-perform-a-quadrant-analysis-in-python-9f84d36f8a24
https://en.wikipedia.org/wiki/Murilo_Bustamante

Github Project Link:

https://github.com/Man-Jason/INFSCI1520-Final-Project